Robust Fitting of Geometric Primitives on LiDAR Data

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Motivation

Computer vision applications [5] for **autonomous driving** are based on digital cameras or 3D sensors. The crucial point of the motion tracking is **object detection**.

Objectives

Our aim is to prove that geometrical primitives can be detected **rapidly** on LiDAR data. The reconstructed objects are **planes**, **spheres** and **cylinders** with robust surface fitting algorithms.

Proposed Solution

Each novel method for object detection includes a model fitting process on LiDAR data. Our approach copes well with special properties of LiDAR scans, like **self-occlusion** and **varying sparsity of the points**.

Object detection

Our shape detection based on RANSAC [4] (RAndom SAmpling Consensus) algorithm. It fits a model from randomly sampled minimal subsets of an observed dataset iteratively. RANSAC requires **3** points for plane and cylinder, and **4** points for sphere fitting.

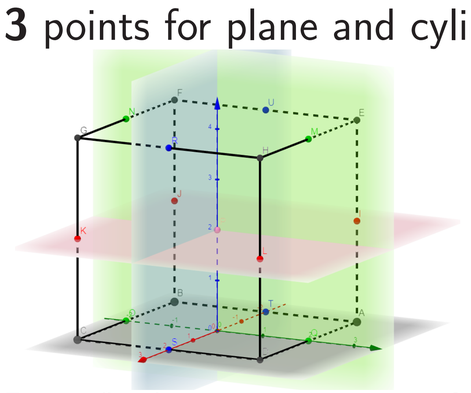


Figure: Synthetic point generation of modified RANSAC.

The new method deals with one chosen point P. The estimation of the plane uses synthetic points with a distance t along coordinate planes. The closest real points to the synthetic points give the input set of the model fitting.

Cylinder and sphere detection uses every point of the cloud in t distance to the point P as inlier. The minimal size of the point set is denoted by k.

The plane [2], sphere [3] and cylinder [1] fitting algorithms are based on least-squares optimization. **The proposed modification of the sphere approximates the center from geometric calculations** instead of the average of the points because of self-occlusion problems. **The direction vector of the cylinder is determined with iterative interval bisection** of the parameters of the spherical coordinates instead of exhaustive search.

Synthetic tests

Qualitative parameters were synthetically tested on spheres and cylinders: noise level from $\mathbf{0.01}$ to $\mathbf{1.00}m$ (red lines), point sparsity from **610** to **15** points (yellow lines) and occlusion ratio from $\mathbf{0}^\circ$ to $\mathbf{357}^\circ$ (blue lines).

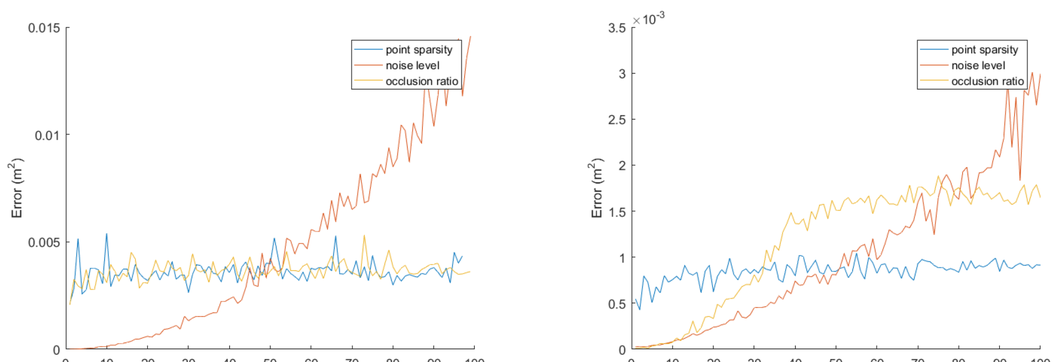


Figure: Error of sphere (right) and cylinder (left) fitting.

Real indoor tests

Indoor tests were used to tune the free parameters of the algorithms. Based on our experiments, the optimal value of t is between 0.3 and 0.5 m. The minimal number of points for the case of spheres and cylinders (k) is **15**. This sample size mitigate the effect of noise.

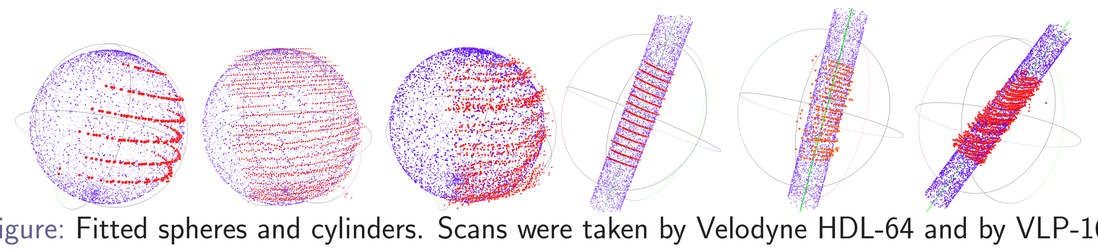


Figure: Fitted spheres and cylinders. Scans were taken by Velodyne HDL-64 and by VLP-16.

Real outdoor tests

Outdoor tests are taken to test the plane and cylinder fitting algorithms. The urban environment is rich in planar surfaces: ground, building facades, etc. Cylinders also appear near roads: advertising pillars and lamp posts can be modeled as cylinders.

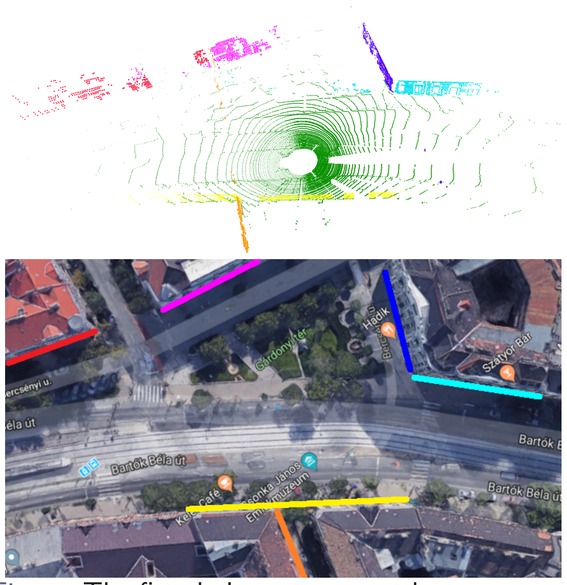


Figure: The fitted planes to an outdoor measurement and the map of the real environment.

	N	$T_{original}$	$T_{proposed}$
1.	52857	**16.434**	17.945
2.	4904	10.323	**9.675**
3.	2239	8.932	**7.830**
4.	2101	7.525	**7.333**
5.	1866	7.292	**6.752**
6.	1518	7.043	**6.425**
7.	1344	6.992	**6.182**

Table: Time demand of the plane detectors. $T_{original}$ and $T_{proposed}$ denote the original RANSAC and the proposed algorithm. Test results are visualized in the left figure. Tests were run **1000**-times and average values are listed in second.

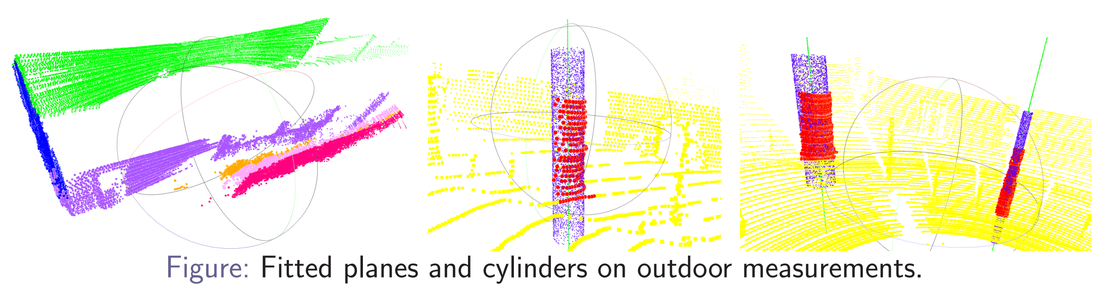


Figure: Fitted planes and cylinders on outdoor measurements.

Conclusion

The proposed fitting algorithms are fast and efficient. The modified RANSAC **needs fewer iterations, decreasing the time demand**. The self-occlusion and sparsity are easy to handle by our novel methods. The algorithms are only limited by high levels of noise in the point cloud. Synthetic, real indoor and outdoor tests confirm our results.

Acknowledgements

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References

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